CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 9: Triangles (Congruency in Triangles)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Two angles of a triangle are 48° and 67°. Find the third angle.

2. An exterior angle of a triangle is 110° and one interior opposite angle is 45°. Find the other two angles of the triangle.

3. Which condition proves congruency: $AB = DE$, $BC = EF$, $\angle B = \angle E$?

4. Which condition proves congruency: $\angle A = \angle D$, $\angle C = \angle F$, $AC = DF$?

5. $\triangle ABC \cong \triangle RQP$. Which side equals AB and which angle equals $\angle C$?

6. Can two triangles with angles $50^\circ, 60^\circ, 70^\circ$ each be said to be congruent?

7. In triangle ABC, AD is perpendicular to BC and $BD = DC$. Prove that $AB = AC$.

8. $\triangle ABC \cong \triangle DEF$, $BC = 2x - 3$, $EF = 11$, $\angle A = 55^\circ$, $\angle D = (y + 15)^\circ$. Find x and y.

9. In right triangles PQR and XYZ, $\angle Q = \angle Y = 90^\circ$, $PR = XZ$ and $PQ = XY$. Name the condition and the angle equal to $\angle R$.

10. A point P is 5 cm from one end of a segment AB and lies on its perpendicular bisector. How far is P from the other end?

Answer Key

1. 65°
2. 65° and 70°
3. S.A.S.
4. A.S.A.
5. $AB = RQ$ and $\angle C = \angle P$
6. Not necessarily; equal angles do not fix the size
7. $\triangle ADB \cong \triangle ADC$ (S.A.S.: $BD = DC$, $\angle ADB = \angle ADC = 90^\circ$, AD common), so $AB = AC$
8. $x = 7$, $y = 40$
9. R.H.S.; $\angle Z$
10. 5 cm

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